

Dr. Babasaheb Ambedkar Technological University, Lonere
(State Technological University)

Summer Examination May 2019

B. Tech. Course in Electronics and Telecommunication Engineering

Subject: Satellite Communication [BET 804] Semester: VIII

Date: 14.05.2019

Time: 3Hours

Max Marks: 70

Instruction:

1. Question numbers 1 is compulsory and carry 10 marks whereas question number 2 to question number 7 carries 12 marks each.
2. Attempt any five questions from question number 2 to question number 7.
3. Illustrate your answers with neat sketches, diagram etc. Wherever necessary.
4. Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that part is part of the examination.
5. If some part or parameters is notice to be missing, you may appropriately assume it and should mention it clearly.

Que. 1 Explain the following terms in short (one line) (10)

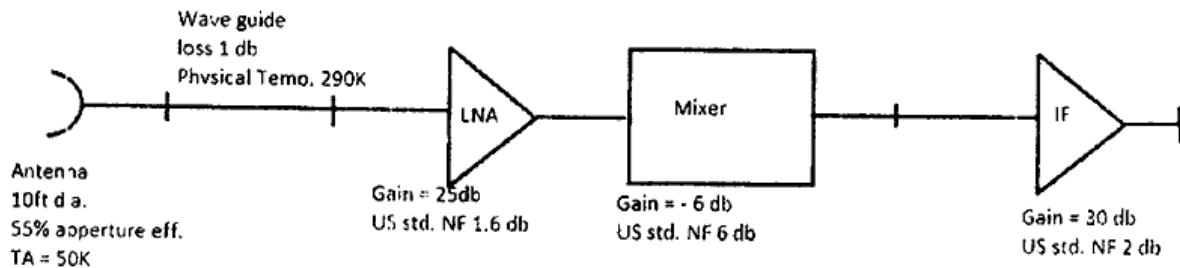
- a. Despun
- b. Aperture
- c. Attitude
- d. Input and Output Back off
- e. Burst
- f. Anomaly
- g. Hopping
- h. Meridian
- i. Sub Satellite points
- j. Transponder

Que. 2 Explain the look angle and prove that

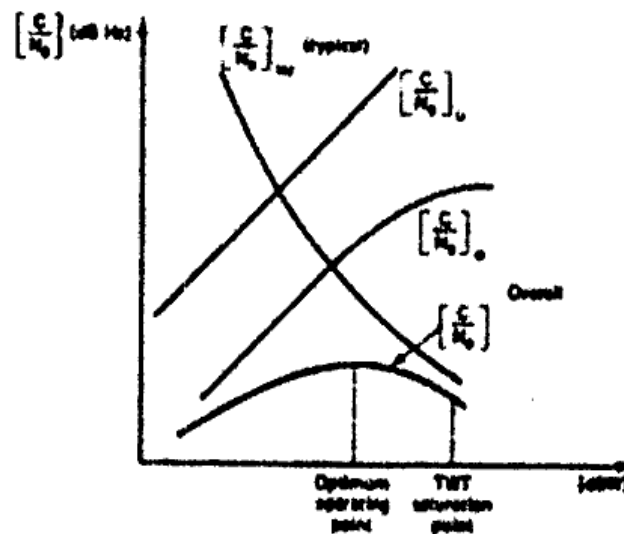
$$\cos(El) = \frac{\sin(\gamma)}{[1 + \left(\frac{r_e}{r_s}\right)^2 - 2\left(\frac{r_e}{r_s}\right)\cos(\gamma)]^{\frac{1}{2}}}$$

Que. 3 What is Transponder? Explain the block diagram of the transponder arrangement with frequency distribution. Also explain the single conversion and double conversion transponder.

Que. 4 A receiver operating at 2800 MHz is shown in block diagram. Calculate its G/T in dBK^{-1} referred to the output port of the antenna.



Que. 5 Prove and explain following relationship with respect to overall C/N ratio



Que. 6 Assume that TDMA system uses a 125 μs frame time. Find the number of channels / earth station when:

- No time is lost in overhead, preamble and the like.
- A 5 μs preamble is added to the beginning of each earth station transmission
- A 5 μs preamble is added to each frame and 2 μs guard band is allowed between each transmission.

Que. 7 Explain Satellite Mobile Communication.